

Complete Artificial Intelligence Fundamentals Mastery Syllabus

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Program Overview

This syllabus is designed to build strong foundational and practical expertise in Artificial Intelligence (AI), preparing learners for careers in:

- Artificial Intelligence Engineering
- Machine Learning Engineering
- Data Science
- NLP Engineering
- Computer Vision Engineering
- AI Research
- Generative AI Development
- AI Product Engineering

The syllabus includes:

- AI Foundations
 - Mathematics for AI
 - Python Programming
 - Machine Learning Basics
 - Deep Learning Foundations
 - Natural Language Processing
 - Computer Vision
 - Generative AI
 - AI Ethics
 - AI Deployment Basics
 - Real-time AI Projects
 - Interview Preparation
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Phase 1 — Introduction to Artificial Intelligence

Module 1: Fundamentals of AI

Topics

- What is Artificial Intelligence?
- History of AI
- Evolution of AI
- AI vs ML vs DL vs Data Science
- Types of AI
- Narrow AI
- General AI
- Super AI
- Applications of AI
- Real-world AI Systems
- AI Lifecycle
- Future of AI

Hands-on

- Exploring AI tools and platforms
 - Understanding AI workflows
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Module 2: AI Domains & Applications

Topics

- Machine Learning
- Deep Learning
- Natural Language Processing
- Computer Vision
- Robotics
- Expert Systems
- Recommendation Systems
- Generative AI
- Conversational AI
- Autonomous Systems

Industry Use Cases

- Healthcare
- Banking
- Retail
- Education

- Cybersecurity
 - Manufacturing
 - Transportation
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Phase 2 — Mathematics for AI

Module 3: Mathematics Foundations

Topics

- Basic Algebra
- Linear Algebra
- Matrices
- Vectors
- Probability
- Statistics
- Calculus Basics
- Derivatives
- Optimization

Important Concepts

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation
- Covariance
- Probability Distributions

Hands-on

- Mathematical calculations using Python
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Phase 3 — Python Programming for AI

Module 4: Python Fundamentals

Topics

- Python Installation
- Variables

- Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- OOP Basics
- Exception Handling
- File Handling

Hands-on

- Python mini projects
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Module 5: Python Libraries for AI

Libraries

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Topics

- Data Handling
- Numerical Computing
- Visualization
- Basic ML Utilities

Hands-on

- Data analysis mini projects
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Phase 4 — Data Processing & Visualization

Module 6: Data Preprocessing

Topics

- Data Cleaning
- Handling Missing Values
- Outlier Detection
- Feature Scaling

- Encoding Techniques
- Data Transformation
- Train-Test Split

Hands-on

- End-to-end preprocessing pipelines
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Module 7: Data Visualization

Topics

- Visualization Principles
- Line Charts
- Bar Charts
- Histograms
- Scatter Plots
- Heatmaps
- Dashboard Concepts

Hands-on

- Interactive data dashboards
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Phase 5 — Introduction to Machine Learning

Module 8: Machine Learning Basics

Topics

- What is Machine Learning?
- ML Workflow
- Types of ML
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Training & Testing Data
- Model Evaluation
- Bias & Variance

Hands-on

- Building first ML models
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Module 9: Supervised Learning

Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machine
- KNN

Metrics

- Accuracy
- Precision
- Recall
- F1 Score

Projects

- Student Performance Prediction
 - House Price Prediction
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Module 10: Unsupervised Learning

Algorithms

- K-Means Clustering
- Hierarchical Clustering
- PCA

Projects

- Customer Segmentation
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Phase 6 — Deep Learning Foundations

Module 11: Neural Networks

Topics

- Artificial Neurons
- Perceptrons
- Neural Networks

- Activation Functions
- Forward Propagation
- Backpropagation
- Loss Functions
- Optimizers

Frameworks

- TensorFlow
- Keras

Hands-on

- Building simple neural networks
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Module 12: Introduction to Deep Learning

Topics

- Deep Learning Basics
- CNN Fundamentals
- RNN Basics
- LSTM Introduction
- Transfer Learning

Projects

- Handwritten Digit Recognition
 - Image Classification
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Phase 7 — Natural Language Processing

Module 13: NLP Fundamentals

Topics

- Text Processing
- Tokenization
- Stopwords
- Stemming
- Lemmatization
- TF-IDF
- Word Embeddings
- Transformers Basics

Projects

- Sentiment Analysis
 - Text Classification
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Phase 8 — Computer Vision

Module 14: Computer Vision Basics

Topics

- Image Processing
- OpenCV Basics
- CNN Architecture
- Object Detection Basics
- Face Detection

Projects

- Face Recognition
 - Image Classifier
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Phase 9 — Generative AI

Module 15: Introduction to Generative AI

Topics

- What is Generative AI?
- Large Language Models
- Prompt Engineering
- AI Chatbots
- Image Generation Basics
- AI Assistants
- RAG Concepts
- AI Agents Basics

Models

- GPT
- Gemini
- Claude
- Llama

Projects

- AI Resume Generator
 - AI Chatbot
 - AI Content Generator
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Phase 10 — AI Ethics & Responsible AI

Module 16: Responsible AI

Topics

- AI Ethics
- Bias in AI
- Fairness
- Explainable AI
- Privacy & Security
- Responsible AI Development
- AI Regulations

Case Studies

- Ethical AI applications
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Phase 11 — AI Deployment Basics

Module 17: Deploying AI Models

Topics

- Model Serialization
- Flask APIs
- FastAPI Basics
- Docker Basics
- Cloud Deployment Overview

Hands-on

- Deploying AI applications
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Phase 12 — AI Tools & Platforms

Module 18: AI Ecosystem

Tools & Platforms

- Google Colab
- Jupyter Notebook
- Kaggle
- Hugging Face
- OpenAI APIs
- TensorFlow Hub
- GitHub for AI

Hands-on

- AI project collaboration
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Phase 13 — Real-Time AI Projects

Beginner Projects

- Calculator AI Assistant
- Student Result Prediction
- Weather Prediction Model
- Basic Chatbot

Intermediate Projects

- Sentiment Analysis System
- Face Detection System
- Recommendation System
- AI-powered Dashboard

Advanced Projects

- Generative AI Assistant
 - Intelligent Resume Analyzer
 - AI Interview Bot
 - AI Healthcare Prediction System
 - Real-time Traffic Prediction System
 - Smart Educational AI Platform
 - Enterprise AI Knowledge Assistant
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Phase 14 — Interview Preparation

Module 19: AI Interview Preparation

Topics

- Python Questions
- AI Fundamentals Questions
- ML Questions
- Statistics Questions
- NLP Questions
- Deep Learning Questions
- Case Studies
- AI Problem Solving

Hands-on

- Mock interviews
 - AI coding challenges
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Recommended Tools

Programming & Development

- Python
- Jupyter Notebook
- VS Code

AI Frameworks

- TensorFlow
- Keras
- PyTorch Basics
- Scikit-learn

Visualization

- Matplotlib
- Seaborn
- Plotly

AI Platforms

- Hugging Face

- Kaggle
 - Google Colab
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Recommended Learning Timeline

Beginner Level (1-2 Months)

- AI Fundamentals
- Python Basics
- Mathematics
- Data Handling

Intermediate Level (3-5 Months)

- Machine Learning
- NLP
- Computer Vision
- Deep Learning Basics

Advanced Level (6-12 Months)

- Generative AI
 - Deployment
 - AI Research
 - Enterprise AI Projects
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Certifications

Recommended Certifications

- Microsoft AI Fundamentals
 - Google AI Certification
 - IBM AI Engineering Certificate
 - TensorFlow Developer Certificate
 - AWS AI Practitioner
 - Oracle AI Foundations Certification
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Outcome After Completion

After completing this syllabus, learners can become:

- AI Engineer
 - ML Engineer
 - Data Scientist
 - NLP Engineer
 - AI Product Developer
 - AI Research Assistant
 - Computer Vision Engineer
 - Generative AI Developer
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Best Learning Strategy

Daily Routine

- 2 Hours Theory
- 3 Hours Coding Practice
- 1 Hour Mathematics
- 1 Hour AI Project Building

Weekly Routine

- Build one mini AI project
 - Practice Kaggle datasets
 - Read AI blogs and papers
 - Upload projects to GitHub
 - Explore new AI tools
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Final Capstone Projects

Enterprise-Level Projects

- AI-powered Recruitment System
- Enterprise AI Chatbot
- AI Learning Assistant Platform
- Real-time Fraud Detection System
- Smart Traffic Prediction Platform
- AI Healthcare Diagnostic System
- Generative AI Content Platform

- Intelligent Business Analytics Assistant
 - AI Security Monitoring System
 - Autonomous AI Agent Platform
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Final Advice

To become successful in Artificial Intelligence:

1. Master mathematics and logic
2. Learn Python deeply
3. Build real-world AI projects
4. Focus on data preprocessing
5. Practice machine learning consistently
6. Learn deployment and APIs
7. Understand AI ethics
8. Stay updated with Generative AI trends
9. Read research papers regularly
10. Build strong GitHub portfolios